

# Dog Shelter Management System

## Design Specification

Fernando Camberos

Kwane Galang

Korapat Jaengsrivong

Andy Lim

Fredy Lung

May 2026

## Contents

|          |                                             |           |
|----------|---------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                         | <b>4</b>  |
| 1.1      | Purpose . . . . .                           | 4         |
| 1.2      | Intended Audience . . . . .                 | 4         |
| 1.3      | System Overview . . . . .                   | 4         |
| 1.4      | Scope . . . . .                             | 4         |
| <b>2</b> | <b>System Overview and Navigation</b>       | <b>4</b>  |
| <b>3</b> | <b>Page and Feature Breakdown</b>           | <b>4</b>  |
| 3.1      | Home / Dog Listing Page . . . . .           | 4         |
| 3.2      | Dog Profile Page . . . . .                  | 5         |
| 3.3      | Dog Intake Page . . . . .                   | 6         |
| 3.4      | Dog Status Tracking . . . . .               | 7         |
| 3.5      | Medical Records Page . . . . .              | 7         |
| 3.6      | Adoption Application Page . . . . .         | 8         |
| 3.7      | Application Review Page . . . . .           | 9         |
| 3.8      | Admin Dashboard . . . . .                   | 10        |
| <b>4</b> | <b>Workflow Overview</b>                    | <b>11</b> |
| <b>5</b> | <b>Workflow Diagram</b>                     | <b>12</b> |
| <b>6</b> | <b>System Dependencies and Technologies</b> | <b>12</b> |
| 6.1      | Frontend . . . . .                          | 12        |
| 6.2      | Backend . . . . .                           | 13        |
| 6.3      | Database . . . . .                          | 13        |

|                                 |           |
|---------------------------------|-----------|
| 6.4 Development Tools . . . . . | 13        |
| <b>7 Future Work</b>            | <b>13</b> |
| <b>8 Glossary</b>               | <b>14</b> |
| <b>9 References</b>             | <b>14</b> |

## Revision History

| Revision # | Change Description                                             | Date       | Revised By | Approved By |
|------------|----------------------------------------------------------------|------------|------------|-------------|
| 1.0        | Initial Design Specification including core pages and features | 05/06/2026 |            |             |
| 1.1        | Added Dog Intake and Status Tracking pages and interactions    | 05/07/2026 |            |             |
| 1.2        | Added Medical Records, Adoption System, and Dashboard pages    | 05/08/2026 |            |             |
| 1.3        | Final refinements and future work additions                    | 05/09/2026 |            |             |

# 1 Introduction

## 1.1 Purpose

This Design Specification document describes the structure and layout of the Dog Shelter Management System. It provides a detailed breakdown of system pages, features, and user interactions.

## 1.2 Intended Audience

This document is intended for developers, project managers, and stakeholders who need a clear understanding of the system layout and functionality.

## 1.3 System Overview

The system is a web-based application designed to manage shelter operations including dog intake, medical tracking, adoption processing, and status management. It supports both public users and staff users.

## 1.4 Scope

This document defines the system's user interface structure and feature breakdown. It complements the SRS and SDD by focusing on how features are presented and used within the system.

# 2 System Overview and Navigation

The system consists of multiple pages accessible based on user roles:

- Public Users: Can view dogs and submit adoption applications
- Staff/Admin: Can manage dogs, medical records, and applications

Navigation is handled through a web-based interface where users interact with pages that communicate with the backend API and database.

# 3 Page and Feature Breakdown

## 3.1 Home / Dog Listing Page

Displays all dogs with their current status (Available, Pending, Adopted).

### Features:

- View list of dogs
- Filter by status



- Search dogs
- Navigate to Dog Profile

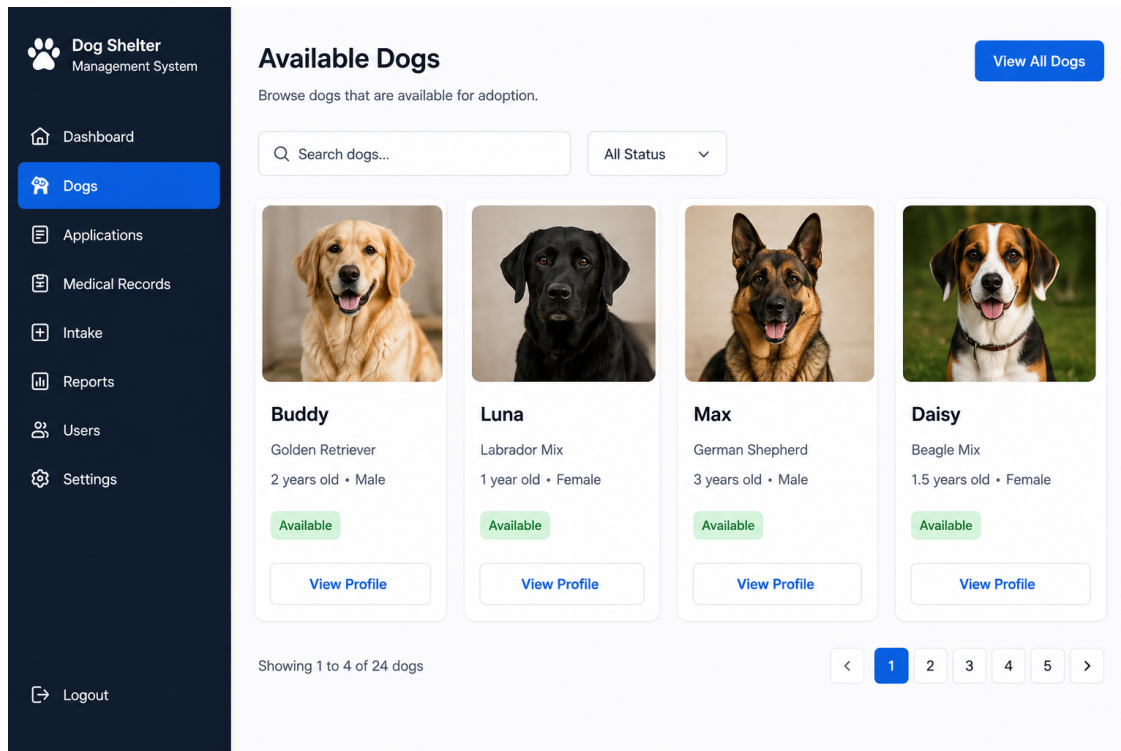


Figure 1: Dog Listing Page displaying available dogs with filtering and search functionality.

## 3.2 Dog Profile Page

Displays detailed information for a selected dog.

### Features:

- View dog information (name, breed, age)
- View current status
- View medical records
- View adoption applications (staff only)

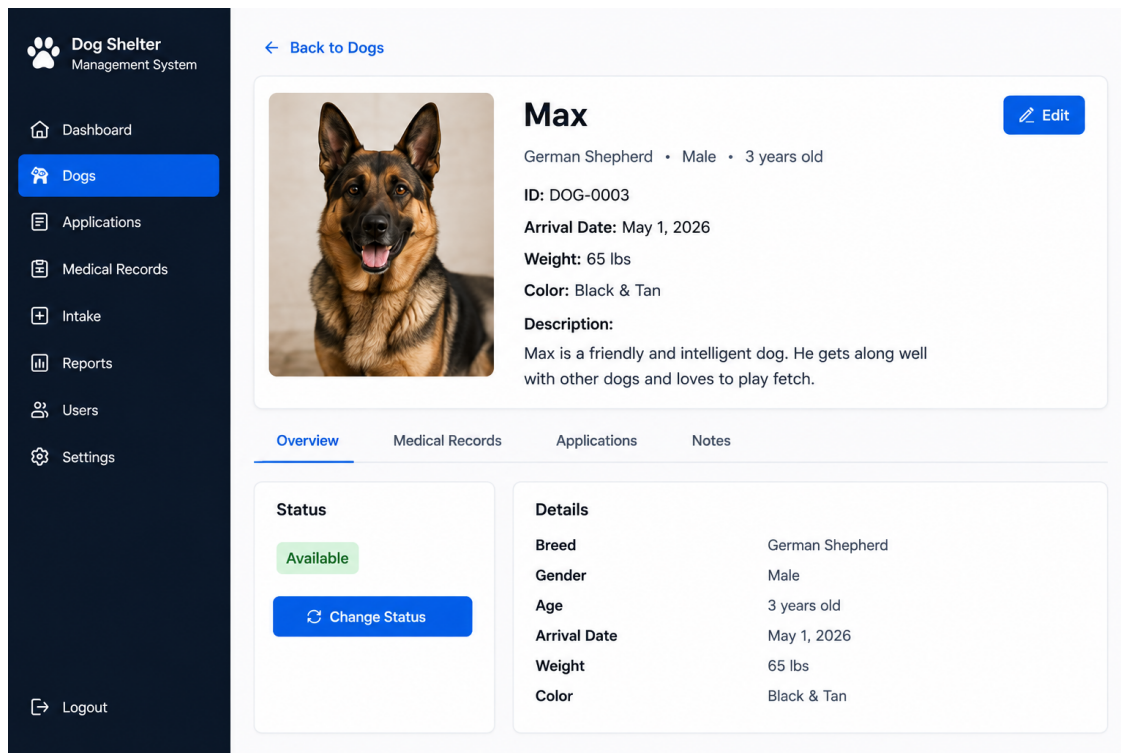


Figure 2: Dog Profile Page showing detailed information, status, and related records for a selected dog.

### 3.3 Dog Intake Page

Allows staff to add new dogs to the system.

#### Inputs:

- Name, Breed, Age, Sex
- Arrival Date
- Intake Notes
- Initial Status

#### Behavior:

- Creates a new dog record
- Stores data in the database
- Displays dog in listing page

**Dog Shelter Management System**

← [Back to Intake](#)

## Add New Dog

Enter the information for the new dog.

**Name \***

**Color**

**Breed \***

**Weight (lbs)**

**Age \***

**Initial Status \***  

Available ▼

**Sex \***  

Select sex ▼

**Arrival Date \***  

05/15/2026

**Intake Notes**  

Enter any notes about the dog

[Save Dog](#)

Figure 3: Dog Intake Page allowing staff to enter and store new dog information.

### 3.4 Dog Status Tracking

Tracks and updates dog availability.

**Statuses:**

- Available
- Pending
- Adopted

**Behavior:**

- Updated by staff or system actions
- Controls visibility on public pages

### 3.5 Medical Records Page

Allows staff to manage health data for dogs.

**Features:**

- Add vaccination records
- Update health status

- View medical history

**Medical Records - Max**

View and manage medical records for Max.

| Date       | Type        | Description            | Vet / Staff | Next Due   | Actions                                     |
|------------|-------------|------------------------|-------------|------------|---------------------------------------------|
| 05/10/2026 | Vaccination | Rabies Vaccine         | Dr. Smith   | 05/10/2027 | <a href="#">Edit</a> <a href="#">Delete</a> |
| 05/10/2026 | Vaccination | DHPP Vaccine           | Dr. Smith   | 05/10/2027 | <a href="#">Edit</a> <a href="#">Delete</a> |
| 05/05/2026 | Treatment   | Flea & Tick Prevention | Staff User  | 06/05/2026 | <a href="#">Edit</a> <a href="#">Delete</a> |
| 05/01/2026 | Checkup     | Initial Health Check   | Dr. Smith   | -          | <a href="#">Edit</a> <a href="#">Delete</a> |

**Health Summary**

**Overall Health:** Good

**Notes:** Max is in good health. Up to date on vaccinations.

Figure 4: Medical Records Page displaying vaccination history, treatments, and health status updates.

### 3.6 Adoption Application Page

Allows users to submit applications.

#### Inputs:

- Applicant name
- Contact information
- Address
- Pet experience
- Selected dog

#### Behavior:

- Stores application in database
- Sets status to Pending

The screenshot shows a web application interface for a dog shelter. On the left is a dark blue sidebar with a paw print icon and the text 'Dog Shelter Management System'. Below this are menu items: 'Dashboard', 'Dogs', 'Applications' (highlighted in blue), 'Medical Records', 'Intake', 'Reports', 'Users', 'Settings', and 'Logout'. The main content area has a light gray background. At the top left of this area is a link '← Back to Dogs'. The title 'Adoption Application' is centered, followed by the text 'You are applying to adopt: Max'. Below this is a form titled 'Applicant Information'. The form contains several fields: 'Full Name \*' (text input), 'Phone Number \*' (text input), 'Email \*' (text input), 'Address \*' (text input), 'City \*' (text input), 'Zip Code \*' (text input), 'Pet Experience \*' (a dropdown menu with 'Select experience' and a downward arrow), and 'Why do you want to adopt?' (a larger text area with the placeholder 'Tell us why you want to adopt this dog...'). A green button labeled 'Submit Application' is at the bottom right of the form.

Figure 5: Adoption Application Page where users submit requests to adopt a dog.

### 3.7 Application Review Page

Allows staff to review applications.

**Actions:**

- Approve application
- Reject application
- Add notes

**Behavior:**

- Updates application status
- Updates dog status accordingly



**Dog Shelter Management System**

← [Back to Applications](#)

**Application Review**

Review and manage adoption applications.

Application ID: APP-2026-0017

[Pending Review](#)

**Sarah Johnson**  
sarah.johnson@email.com  
(555) 123-4567

Submitted: May 15, 2026

Dog: [Max \(DOG-0003\)](#)

Application Status: [Pending Review](#)

**Application Details**

Full Name: Sarah Johnson  
Email: sarah.johnson@email.com  
Phone: (555) 123-4567  
Address: 123 Main Street, Austin, TX 78701  
Pet Experience: Experienced

**Why do you want to adopt?**  
I have always loved dogs and currently have a big yard and flexible schedule. I believe Max would be a great addition to my family.

**Additional Notes:**  
I have previous experience with German Shepherds.

**Questionnaire Responses**

Do you own or rent your home?  
Own

Do you have a fenced yard?  
Yes

Do you have other pets?  
No

Who will be the primary caregiver?  
Me

How many hours will the dog be alone during the day?  
2-3 hours

[View Application](#) [Reject Application](#) [Approve Application](#)

Figure 6: Application Review Page allowing staff to evaluate and manage adoption applications.

### 3.8 Admin Dashboard

Central management page for staff.

#### Features:

- View total dogs
- View dogs by status
- View pending applications
- Manage system data



Figure 7: Admin Dashboard providing an overview of system data, including dog counts, applications, and activity.

## 4 Workflow Overview

The system follows a standard interaction flow:

1. Staff adds a dog using the Dog Intake Page
2. Dog is stored in the database and appears as Available
3. Users view dogs and submit adoption applications
4. Staff reviews applications
5. Approved applications change dog status to Pending
6. Final adoption updates status to Adopted
7. Dashboard reflects all updates

## 5 Workflow Diagram

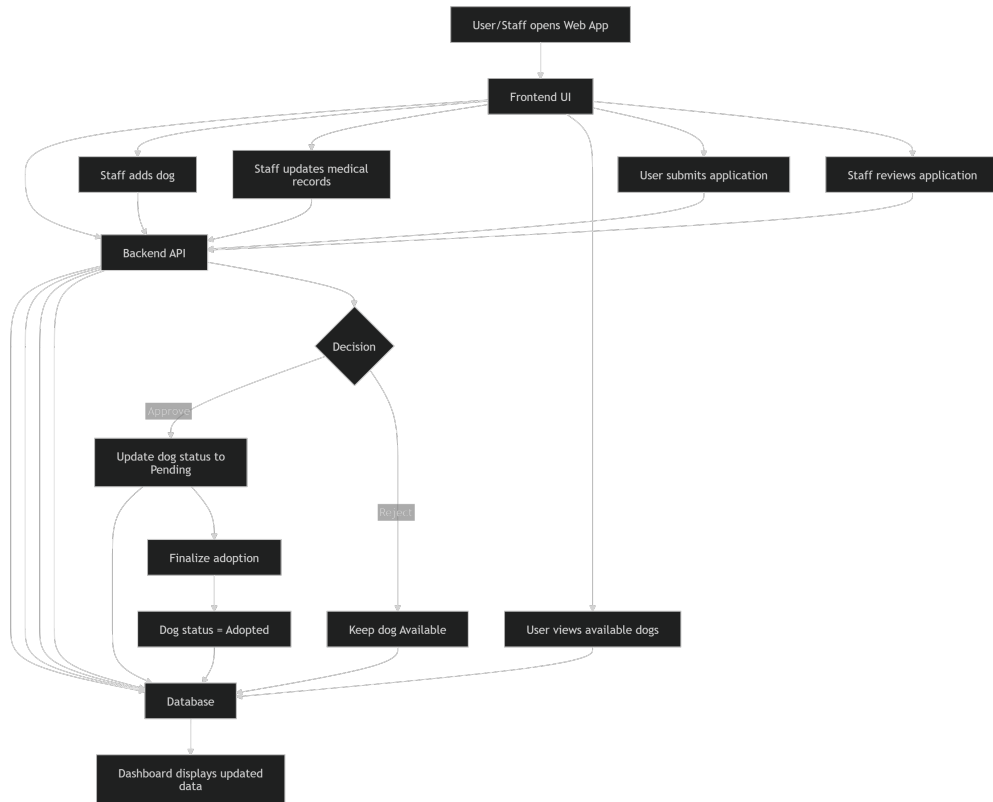


Figure 8: High-level workflow diagram of the Dog Shelter Management System showing interaction between users, frontend, backend API, and database.

As shown in Figure 8, the system processes user and staff interactions through a web-based frontend interface. Requests are sent to the backend API, which handles business logic and communicates with the database to store and retrieve information. The workflow ensures that dog intake, medical updates, and adoption processes are consistently managed across the system.

## 6 System Dependencies and Technologies

The Dog Shelter Management System relies on the following technologies and dependencies to function within a Dockerized environment.

### 6.1 Frontend

- React.js
- react-dom



- react-scripts
- Axios (API communication)
- Bootstrap or Material UI (UI components)

## 6.2 Backend

- Node.js
- Express.js
- body-parser
- cors
- dotenv
- jsonwebtoken (authentication)

## 6.3 Database

- MySQL
- mysql2 (Node.js connector)

## 6.4 Development Tools

- Docker
- Docker Compose
- GitHub (version control)

# 7 Future Work

Potential future enhancements include:

- Email notifications for application status
- Image uploads for dog profiles
- Advanced search and filtering
- Appointment scheduling for adoption visits

## 8 Glossary

| Term        | Definition                                           |
|-------------|------------------------------------------------------|
| Intake      | Process of adding a new dog                          |
| Status      | Current state of a dog (Available, Pending, Adopted) |
| Application | Request submitted by a user to adopt a dog           |
| Dashboard   | Admin interface for managing system data             |

## 9 References

- SRS Document
- SDD Document
- CS3338 Project Guidelines